

# KINGSHUK CHATTERJEE

Software Developer | ML Engineer  
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## EDUCATION

**KIIT University**  
Bachelor of Technology in Computer Science  
Research Member, KIIT Research & Software Club (KRSC)

Bhubaneswar, Odisha  
2022 – 2026

## SKILLS

**Languages** Python, JavaScript, Rust, C, C++, SQL  
**Frameworks & Libraries** React, Node.js, Flask, FastAPI, Flutter, PyTorch, TensorFlow, OpenCV, CleanRL  
**Databases & BaaS** MongoDB, PostgreSQL, Supabase, Firebase  
**Developer Tools** Git, REST APIs, Figma, Blender  
**Machine Learning** PINNs, CNNs, LSTMs, Time-Series Modeling, Feature Engineering  
**Systems & GPU** AVX2 SIMD, OpenAI Triton, CUDA Kernels, ctypes C++ Bridge

## EXPERIENCE

**Full Stack Developer (Part-time)** — Nexveda Technologies  
Developed a full-stack education platform using React.js, Python, MongoDB, and PostgreSQL; engineered RESTful APIs for scalable frontend-backend integration.

Jan 2026 – Present

**Frontend Developer (Freelance)** — Ionic Design  
Delivered responsive web interfaces using HTML, CSS, and Python-based tooling.

Apr 2026

**Intern** — Indian Oil Corporation Ltd (IOCL)  
Worked with industrial networking infrastructure and Cisco protocols.

Nov 2024 – Jan 2025

## SELECTED PROJECTS

- **OmniRay : AVX2-Accelerated Deep RL Active SLAM Engine** GitHub
  - Developed **OmniRay**, implementing a custom C++ AVX2 SIMD raycaster with **0.038ms scan times** (79x faster than PyMunk) and a parallelized NumPy particle filter executing steps in under 3.2ms.
  - Deployed a **CNN-MLP PPO agent** achieving **95.1% drift reduction**; governed actions via a closed-loop **5-layer self-adaptive system** with neural reward-tuning and continual learning.
- **Dynamic Gradient PINNs** GitHub
  - Developed a ResSIREN-Fourier PINN with PCGrad and Softplus Physics Anchor; achieved 32,700x forward inference speedup over OpenFOAM with robust parameter discovery under 10% noise.
- **Triton Fuser : Hydrodynamic Order Book PINN** GitHub
  - Modeled limit order book dynamics as a fluid-density PDE and trained a PINN (PyTorch) on live Binance L2 data.
  - Built a C++ AVX2 SIMD ingestion engine (~14ns/tick) and an OpenAI Triton fused GEMM + Softplus kernel for NVIDIA T4 GPUs.
- **Krishi Sahayak : AI Agricultural Platform (Government funded : NIELIT)** Website and Application
  - Improved platform performance by 70% and supported 50+ active beta users; built with Flutter, FastAPI, Supabase, and Firebase.
  - Integrated CNN-based crop disease detection with TFLite edge inference, live mandi prices, weather intelligence, and ML-driven sowing recommendations.
- **Sys Logger (Government funded : NIELIT)** GitHub | Website
  - Architected a production-deployed distributed system via PM2 monitoring 100+ active network endpoints in real time; built responsive dashboards for system activity.
- **FinReport : Earnings Forecast — ICMLDE Conference Paper** GitHub | Sciencedirect
  - Applied NLP sentiment analysis and quantitative financial indicators to build an ML earnings forecasting model; research presented at ICMLDE Conference.
- **Apex OS : Neural-Native Operating System** (research, in development)
  - Developed a bare-metal x86\_64 Rust OS featuring an RL-driven CPU scheduler and a real-time, neural-guided control loop.
  - Implemented TPM 2.0 verified boot, capability-based sandboxing, and a zero-copy UDP swarm protocol for decentralized coordination and custom GUI.